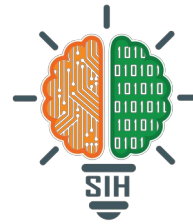


SMART INDIA HACKATHON 2026

AI-Based Intelligent Video Analytics Platform



SMART INDIA
HACKATHON
2026

for Border Surveillance using existing CCTV Infrastructure

Problem Statement ID – Not provided in source

Problem Statement Title – AI-Based Intelligent Video Analytics Platform
for Border Surveillance using existing CCTV Infrastructure

Theme – AI / Computer Vision

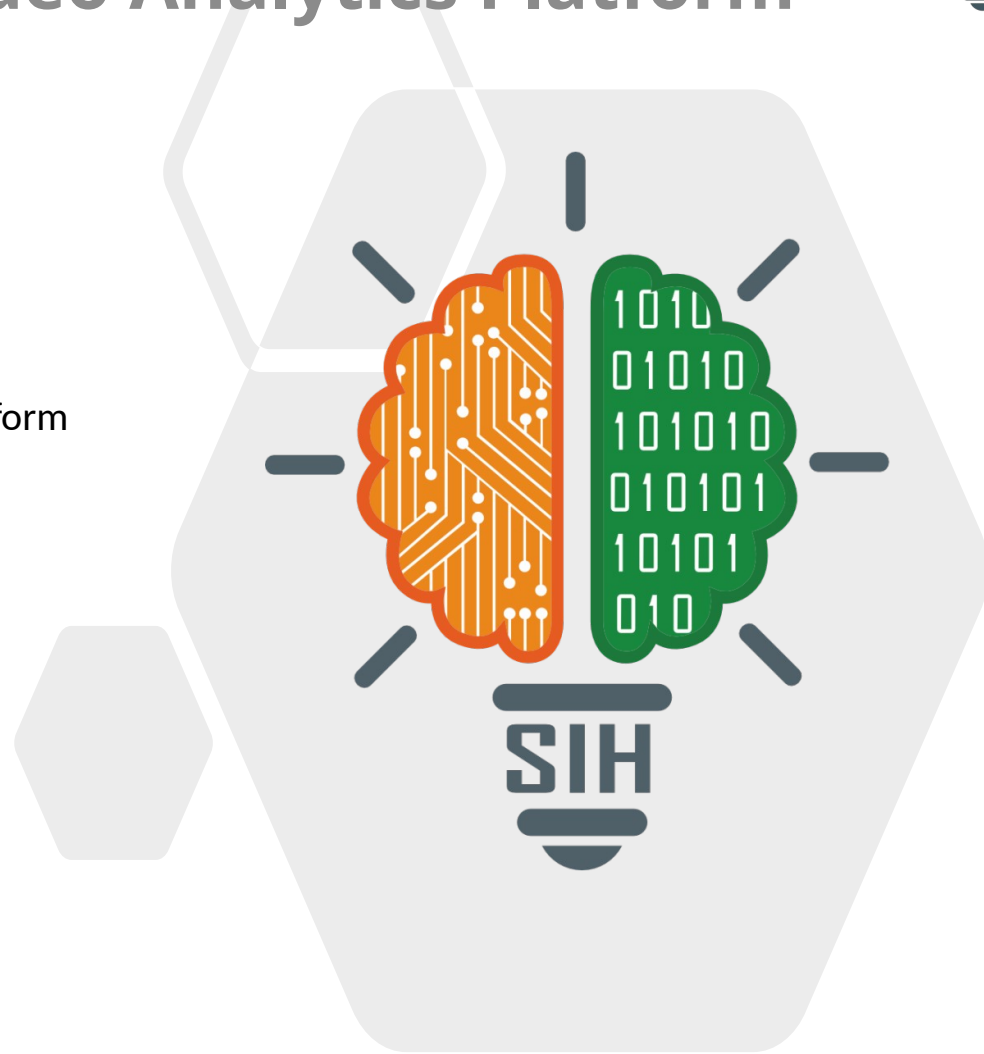
PS Category – Software

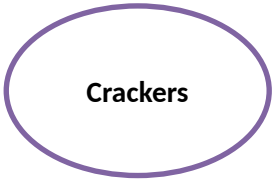
Team ID – Not provided in source

Team Name – Crackers

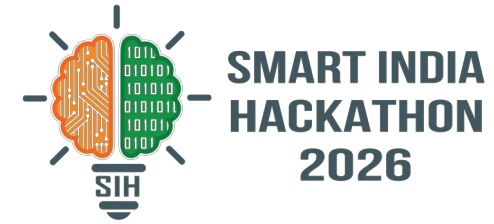
Group Leader – Krish Modi

Team Members – Srushti Patel, Janvi Chhasatiya, Jay Patel, Sumit
Chaudhry, Satyam Singh





IDEA TITLE



❖ Proposed Solution

- Add an AI intelligence layer to existing CCTV / RTSP streams instead of replacing camera infrastructure.
- Detect people, objects and vehicles; track selected events over time; trigger rule-based alerts; store important events.

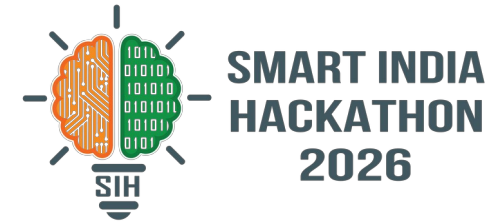
How it addresses the problem

- Reduces dependence on continuous manual monitoring.
- Helps identify restricted-area access, unattended objects and vehicle/plate events in real time.
- Creates event records for later review and audit.

Innovation & uniqueness

- Hybrid pipeline combining pretrained computer-vision models, OCR and rule-based event logic.
- Uses time + distance + state for event decisions rather than relying on a single video frame.
- Designed around existing live CCTV infrastructure.

TECHNICAL APPROACH



- **Technologies / AI Stack**

- YOLO – person, object and vehicle detection.
- OpenCV – video frames, preprocessing and tracking support.
- OCR – reads number-plate text from detected plate regions.
- YuFace – face detection; SFace – face feature extraction / recognition.
- FastAPI – connects AI processing, events and web dashboard; Database – stores event records, timestamps, camera data, alerts and plate records.

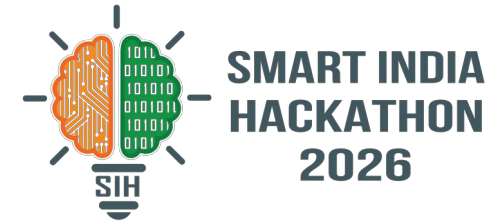
- **Methodology / Process**

- Existing CCTV / RTSP → Frame processing → AI detection → Rule / event analysis → Alert + event record → Dashboard.
- Restricted area: person detected → identity/authorization check → authorized log OR unauthorized alert + snapshot.
- Unattended object: person + bag → owner moves away → object remains stationary → threshold reached → alert.
- Vehicle: vehicle detected → plate cropped → OCR → authorization / entry rule checked.

- **AI Type**

- Hybrid AI: mostly pretrained models + rule-based abnormal-event detection; not a single end-to-end trained model.

FEASIBILITY AND VIABILITY



- **Feasibility**

- Prototype is designed to use existing CCTV / RTSP streams, reducing the need for new camera infrastructure.
- Uses pretrained computer-vision components and a modular FastAPI + database architecture.
- Rules can be adapted to restricted zones, perimeter monitoring and vehicle-entry scenarios.

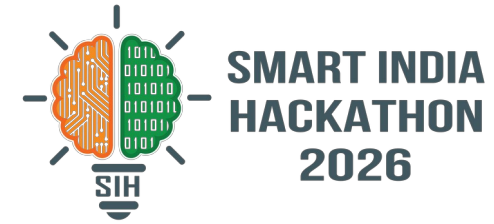
Potential challenges & risks

- Performance depends on camera quality and viewing angle.
- Distant faces and small number plates are harder to recognize.
- Detection rules require tuning for different environments.
- Prototype is demonstrated on selected scenarios.

Strategies / Future improvements

- Improve long-distance recognition and tracking.
- Support multiple live CCTV streams and mobile notifications.
- Add more anomaly / behavior detection and improve border-perimeter deployment testing.

IMPACT AND BENEFITS



- **Potential Impact**

- Supports faster identification of important events in border / perimeter monitoring.
- Helps security personnel focus attention on alerts instead of continuously reviewing every camera feed.
- Provides event history for review and audit.

Benefits

- Real-time visual alerts and optional audio siren for defined events.
- Detection of people, vehicles and objects from existing CCTV input.
- Restricted-area monitoring, unattended-object tracking and vehicle / plate monitoring in one platform.
- Reduces the need for new CCTV infrastructure for the prototype.

Human-Centered Use

- The system is intended to assist security personnel by helping them notice important events faster; it does not replace them.

- **Reference / Research Material**

- Primary source used for this conversion: the provided 1team_Crackers.pdf — AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure.
- The source describes the prototype architecture, AI stack, detection features, event logic, limitations and future improvements.

Technologies Referenced in the Source

- YOLO; OpenCV; OCR; YuFace; SFace; FastAPI; Database; CCTV / RTSP.
- Hybrid AI approach: pretrained computer-vision models + OCR + rule-based event logic.

Reference Links

- No external URLs were provided in the source PDF; therefore no additional links have been added.